

STAT3006 Worksheet 06

1. Real AdaBoost

Real AdaBoost is an additive boosting method that constructs a strong classifier by iteratively adding weak learners with real-valued outputs. Unlike discrete AdaBoost, each weak learner contributes a confidence-weighted prediction.

The additive model takes the form

$$h^{(B)}(\mathbf{x}) = \sum_{b=1}^B \nu f^{(b)}(\mathbf{x}), \quad f^{(b)}(\mathbf{x}) \in \mathbb{R}$$

where $\nu > 0$ is a shrinkage parameter. In this exercise, we set $\nu = 1$ so that

$$h^{(b)}(\mathbf{x}) = \sum_{b=1}^B f^{(b)}(\mathbf{x})$$

The final classifier is

$$\hat{y} = \text{sign}\left(h^{(b)}(\mathbf{x})\right)$$

Real AdaBoost is derived by minimising the exponential loss

$$L(h) = \sum_{i=1}^n \exp\left(-y_i h(x_i)\right), \quad y_i \in \{-1, +1\}$$

(a) Show that minimising the exponential loss after adding a weak learner

$$h^{(b)}(\mathbf{x}) = h^{(b-1)}(\mathbf{x}) + f^{(b)}(\mathbf{x})$$

is equivalent to solving a weighted exponential loss problem:

$$\min_j \sum_{i=1}^n w_i^{(b)} \exp\left(-y_i f(\mathbf{x}_i)\right)$$

where the weights are

$$w_i^{(b)} = \exp\left(-y_i h^{(b-1)}(\mathbf{x}_i)\right)$$

Answer: The loss function is given by

$$\begin{aligned}
L(h) &= \sum_{i=1}^n \exp \left(- y_i h(\mathbf{x}_i) \right) \\
&= \sum_{i=1}^n \exp \left\{ - y_i \left(h^{(b-1)}(\mathbf{x}_i) + f^{(b)}(\mathbf{x}_i) \right) \right\} \\
&= \sum_{i=1}^n \exp \left\{ - y_i h^{(b-1)}(\mathbf{x}_i) \right\} \cdot \exp \left\{ - y_i f^{(b)}(\mathbf{x}_i) \right\}
\end{aligned}$$

Result follows

(b) Assume a weak learner is a tree with Regions $R_1, \dots, R_j$:

$$f^{(b)}(\mathbf{x}) = \gamma_j, \quad \text{if } \mathbf{x} \in R_j$$

Show that the optimal leaf value in Real AdaBoost is

$$\gamma_j = \frac{1}{2} \log \frac{\sum_{i \in R_j, y_i=1} w_i^{(b)}}{\sum_{i \in R_j, y_i=-1} w_i^{(b)}}$$

Answer: First we define the loss function for a given region j ,

$$L_j(h) = \sum_{i=1}^n w_i^{(b)} \exp \left\{ - y_i \gamma_j \right\}$$

We partition L_j into two summations: one where $y_i = -1$ and another one where $y_i = +1$,

$$L_j(h) = \sum_{i \in R_j, y_i=1} w_i^{(b)} \exp(-\gamma_j) + \sum_{i \in R_j, y_i=-1} w_i^{(b)} \exp(\gamma_j)$$

Differentiating with respect to γ_j we get

$$\frac{\partial L_j(h)}{\partial \gamma_j} = - \sum_{i \in R_j, y_i=1} w_i^{(b)} \exp(-\gamma_j) + \sum_{i \in R_j, y_i=-1} w_i^{(b)} \exp(\gamma_j)$$

Letting L.H.S. = 0 we get

$$\begin{aligned}
\sum_{i \in R_j, y_i=1} w_i^{(b)} \exp(-\gamma_j) &= \sum_{i \in R_j, y_i=-1} w_i^{(b)} \exp(\gamma_j) \\
\frac{\exp(\gamma_j)}{\exp(-\gamma_j)} &= \frac{\sum_{i \in R_j, y_i=1} w_i^{(b)}}{\sum_{i \in R_j, y_i=-1} w_i^{(b)}} \\
\exp(2\gamma_j) &= \frac{\sum_{i \in R_j, y_i=1} w_i^{(b)}}{\sum_{i \in R_j, y_i=-1} w_i^{(b)}} \\
2\gamma_j &= \log \frac{\sum_{i \in R_j, y_i=1} w_i^{(b)}}{\sum_{i \in R_j, y_i=-1} w_i^{(b)}} \\
\gamma_j &= \frac{1}{2} \log \frac{\sum_{i \in R_j, y_i=1} w_i^{(b)}}{\sum_{i \in R_j, y_i=-1} w_i^{(b)}}
\end{aligned}$$

(c) In standard boosting, one often writes

$$h^{(b)}(\mathbf{x}) = h^{(b-1)}(\mathbf{x}) + \alpha^{(b)} f^{(b)}(\mathbf{x}) \quad (1)$$

Explain where the step size $\alpha^{(b)}$ appears in Real AdaBoost

Answer: In standard AdaBoost, also called discrete AdaBoost, since $y_i \in \{-1, +1\}$ we have a step size $\alpha^{(b)}$ which transforms the terms in the summation from integer values to real values.

In Real AdaBoost, the predictions aren't discrete and involve both the direction as well as confidence of the prediction. That is, $y \in [0, 1]$. Therefore there isn't an explicit $\alpha^{(b)}$ parameter and is otherwise encoded into the predictor function.

(d) Explain why Real AdaBoost is more flexible than Discrete AdaBoost

Answer: In Discrete AdaBoost, the $\alpha_i^{(b)}$ term is imposed onto all predictions from that learner. Whereas with Real AdaBoost, this term is encoded straight into the predictor as each prediction has both direction and confidence. We can think of it as every single prediction in every learner has an $\alpha^{(b)}$ term of its own, making Real AdaBoost more flexible.